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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 09/22/2008 |
| **PI Name:** | Kaul, Raj K. Ph.D. | | |
| **Project Title:** | Graphite/PE Hybrid Fiber Reinforced Composite with Epoxy and PE Matrices for Cosmic Radiation Shielding | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | RADIATION SHIELDING | | |
| **Program/Discipline--Element/Subdiscipline:** | RADIATION SHIELDING--Radiation shielding | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | NASA CENTER | **Phone:** | 256-544-1084 |
| **Organization Name:** | NASA Marshall Space Flight Center | | |
| **PI Address 1:** | Mail Code ED34 | | |
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| **PI Web Page:** | | | |
| **City:** | Marshall Space Flight Center | **State:** | AL |
| **Zip Code:** | 35812 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-UNSOL |
| **Start Date:** | 05/16/2003 | **End Date:** | 09/30/2007 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 9/30/2007 per PI (8/07) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Penn, Benjamin ( NASA Marshall Space Flight Center )<br>Edwards, David ( NASA Marshall Space Flight Center )<br>Hodge, Andy ( NASA Marshall Space Flight Center )<br>Hulcher, A. ( NASA Marshall Space Flight Center )<br>Trinh, Diep ( NASA Marshall Space Flight Center ) | | |
| **Grant/Contract No.:** | Intramural | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | (1) Perform all necessary theoretical and accelerator-based analyses of the radiation-shielding effectiveness, including fatigue, of these materials<br>(2) Fabricate optimized material design panels for mechanical and physical property testing both before and after radiation exposure. Generate database summarizing results of the study on the effectiveness of the various combinations of material components on both shielding and structural efficiency and make data available to the space radiation community.<br><br>Develop and characterize multi-functional materials that can serve as integral structural members of the space vehicle while providing the necessary radiation shielding for the crew and are potentially, both mission enabling as well as cost effective.<br><br>Identification and characterization of multifunctional materials. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Polyethylene based composites promise multifunctionality for deep-space exploration. Polyethylene based composites demonstrated improved radiation shielding over more typical structural materials in the high-energy and high-charge particle radiation fields associated with the deep space environment. This improvement is a result of the high hydrogen content and lack of heavy elements in the polyethylene. In addition to radiation shielding, the composites have improved mechanical properties compared to pure polyethylene. Furthermore, hypervelocity impact testing representative of micrometeoroid impact and out gas testing demonstrate the suitability of PE composite for use in the space environment. Thus, polyethylene based composite materials can be used as structural materials as well as radiation shielding, leading to the multifunctional performance in space habitats. |
| **Bibliography Type:** | Description: (Last Updated: 02/27/2004) |